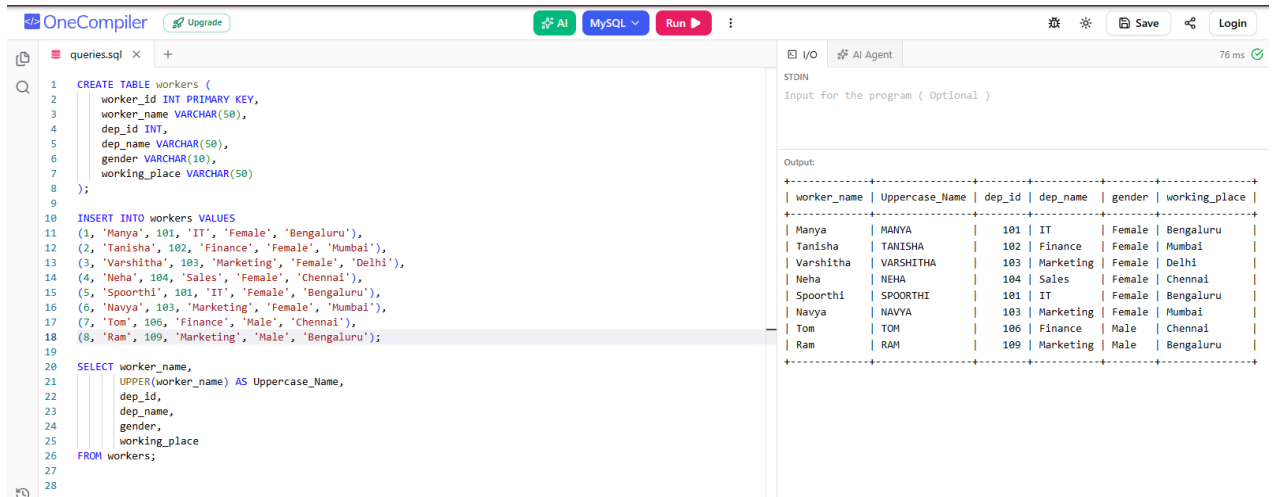


STRING FUNCTION:

❖ UPPER():



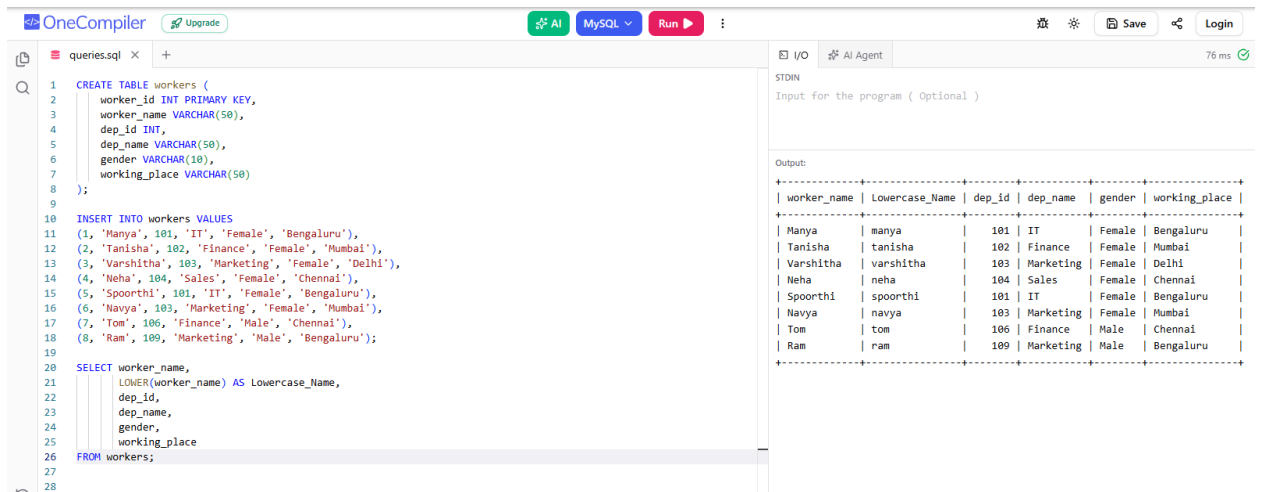
The screenshot shows the OneCompiler MySQL interface. The SQL editor contains the following code:

```
1 CREATE TABLE workers (  
2     worker_id INT PRIMARY KEY,  
3     worker_name VARCHAR(50),  
4     dep_id INT,  
5     dep_name VARCHAR(50),  
6     gender VARCHAR(10),  
7     working_place VARCHAR(50)  
8 );  
9  
10 INSERT INTO workers VALUES  
11 (1, 'Manya', 101, 'IT', 'Female', 'Bengaluru'),  
12 (2, 'Tanisha', 102, 'Finance', 'Female', 'Mumbai'),  
13 (3, 'Varshitha', 103, 'Marketing', 'Female', 'Delhi'),  
14 (4, 'Neha', 104, 'Sales', 'Female', 'Chennai'),  
15 (5, 'Spoorthi', 101, 'IT', 'Female', 'Bengaluru'),  
16 (6, 'Navya', 103, 'Marketing', 'Female', 'Mumbai'),  
17 (7, 'Tom', 106, 'Finance', 'Male', 'Chennai'),  
18 (8, 'Ram', 109, 'Marketing', 'Male', 'Bengaluru');  
19  
20 SELECT worker_name,  
21         UPPER(worker_name) AS Uppercase_Name,  
22         dep_id,  
23         dep_name,  
24         gender,  
25         working_place  
26 FROM workers;  
27  
28
```

The output of the query is displayed on the right:

worker_name	Uppercase_Name	dep_id	dep_name	gender	working_place
Manya	MANYA	101	IT	Female	Bengaluru
Tanisha	TANISHA	102	Finance	Female	Mumbai
Varshitha	VARSHITHA	103	Marketing	Female	Delhi
Neha	NEHA	104	Sales	Female	Chennai
Spoorthi	SPOORTHY	101	IT	Female	Bengaluru
Navya	NAVYA	103	Marketing	Female	Mumbai
Tom	TOM	106	Finance	Male	Chennai
Ram	RAM	109	Marketing	Male	Bengaluru

❖ LOWER():



The screenshot shows the OneCompiler MySQL interface. The SQL editor contains the following code:

```
1 CREATE TABLE workers (  
2     worker_id INT PRIMARY KEY,  
3     worker_name VARCHAR(50),  
4     dep_id INT,  
5     dep_name VARCHAR(50),  
6     gender VARCHAR(10),  
7     working_place VARCHAR(50)  
8 );  
9  
10 INSERT INTO workers VALUES  
11 (1, 'Manya', 101, 'IT', 'Female', 'Bengaluru'),  
12 (2, 'Tanisha', 102, 'Finance', 'Female', 'Mumbai'),  
13 (3, 'Varshitha', 103, 'Marketing', 'Female', 'Delhi'),  
14 (4, 'Neha', 104, 'Sales', 'Female', 'Chennai'),  
15 (5, 'Spoorthi', 101, 'IT', 'Female', 'Bengaluru'),  
16 (6, 'Navya', 103, 'Marketing', 'Female', 'Mumbai'),  
17 (7, 'Tom', 106, 'Finance', 'Male', 'Chennai'),  
18 (8, 'Ram', 109, 'Marketing', 'Male', 'Bengaluru');  
19  
20 SELECT worker_name,  
21         LOWER(worker_name) AS Lowercase_Name,  
22         dep_id,  
23         dep_name,  
24         gender,  
25         working_place  
26 FROM workers;  
27  
28
```

The output of the query is displayed on the right:

worker_name	Lowercase_Name	dep_id	dep_name	gender	working_place
Manya	manya	101	IT	Female	Bengaluru
Tanisha	tanisha	102	Finance	Female	Mumbai
Varshitha	varshitha	103	Marketing	Female	Delhi
Neha	neha	104	Sales	Female	Chennai
Spoorthi	spoorthi	101	IT	Female	Bengaluru
Navya	navya	103	Marketing	Female	Mumbai
Tom	tom	106	Finance	Male	Chennai
Ram	ram	109	Marketing	Male	Bengaluru

❖ LENGTH():

OneCompiler Upgrade

queries.sql

```
1 CREATE TABLE workers (  
2   worker_id INT PRIMARY KEY,  
3   worker_name VARCHAR(50),  
4   dep_id INT,  
5   dep_name VARCHAR(50),  
6   gender VARCHAR(10),  
7   working_place VARCHAR(50)  
8 );  
9  
10 INSERT INTO workers VALUES  
11 (1, 'Manya', 101, 'IT', 'Female', 'Bengaluru'),  
12 (2, 'Tanisha', 102, 'Finance', 'Female', 'Mumbai'),  
13 (3, 'Varshitha', 103, 'Marketing', 'Female', 'Delhi'),  
14 (4, 'Neha', 104, 'Sales', 'Female', 'Chennai'),  
15 (5, 'Spoorthi', 101, 'IT', 'Female', 'Bengaluru'),  
16 (6, 'Navya', 103, 'Marketing', 'Female', 'Mumbai'),  
17 (7, 'Tom', 106, 'Finance', 'Male', 'Chennai'),  
18 (8, 'Ram', 109, 'Marketing', 'Male', 'Bengaluru');  
19  
20 SELECT worker_name,  
21        LENGTH(worker_name) AS Length_Name,  
22        dep_id,  
23        dep_name,  
24        gender,  
25        working_place  
26 FROM workers;  
27  
28
```

Output:

worker_name	Length_Name	dep_id	dep_name	gender	working_place
Manya	5	101	IT	Female	Bengaluru
Tanisha	7	102	Finance	Female	Mumbai
Varshitha	9	103	Marketing	Female	Delhi
Neha	4	104	Sales	Female	Chennai
Spoorthi	8	101	IT	Female	Bengaluru
Navya	5	103	Marketing	Female	Mumbai
Tom	3	106	Finance	Male	Chennai
Ram	3	109	Marketing	Male	Bengaluru

❖ CONCAT():

OneCompiler Upgrade

queries.sql

```
1 CREATE TABLE workers (  
2   worker_id INT PRIMARY KEY,  
3   worker_name VARCHAR(50),  
4   dep_id INT,  
5   dep_name VARCHAR(50),  
6   gender VARCHAR(10),  
7   working_place VARCHAR(50)  
8 );  
9  
10 INSERT INTO workers VALUES  
11 (1, 'Manya', 101, 'IT', 'Female', 'Bengaluru'),  
12 (2, 'Tanisha', 102, 'Finance', 'Female', 'Mumbai'),  
13 (3, 'Varshitha', 103, 'Marketing', 'Female', 'Delhi'),  
14 (4, 'Neha', 104, 'Sales', 'Female', 'Chennai'),  
15 (5, 'Spoorthi', 101, 'IT', 'Female', 'Bengaluru'),  
16 (6, 'Navya', 103, 'Marketing', 'Female', 'Mumbai'),  
17 (7, 'Tom', 106, 'Finance', 'Male', 'Chennai'),  
18 (8, 'Ram', 109, 'Marketing', 'Male', 'Bengaluru');  
19  
20 SELECT worker_name,  
21        working_place,  
22        CONCAT(worker_name, ' - ', working_place) AS Worker_Details  
23 FROM workers;  
24
```

Output:

worker_name	working_place	Worker_Details
Manya	Bengaluru	Manya - Bengaluru
Tanisha	Mumbai	Tanisha - Mumbai
Varshitha	Delhi	Varshitha - Delhi
Neha	Chennai	Neha - Chennai
Spoorthi	Bengaluru	Spoorthi - Bengaluru
Navya	Mumbai	Navya - Mumbai
Tom	Chennai	Tom - Chennai
Ram	Bengaluru	Ram - Bengaluru

❖ SUBSTRING():

OneCompiler Upgrade

queries.sql

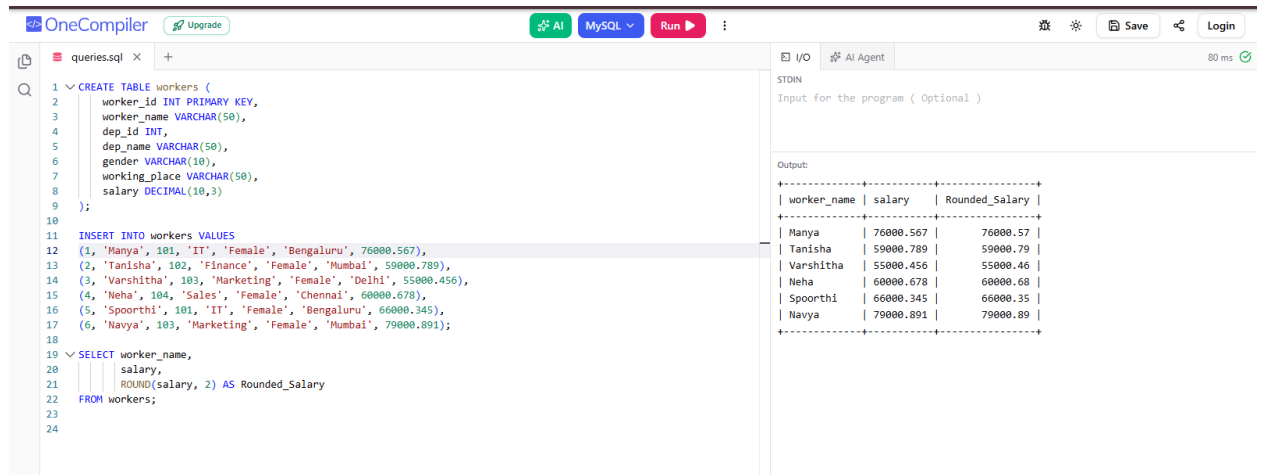
```
1 CREATE TABLE workers (  
2   worker_id INT PRIMARY KEY,  
3   worker_name VARCHAR(50),  
4   dep_id INT,  
5   dep_name VARCHAR(50),  
6   gender VARCHAR(10),  
7   working_place VARCHAR(50)  
8 );  
9  
10 INSERT INTO workers VALUES  
11 (1, 'Manya', 101, 'IT', 'Female', 'Bengaluru'),  
12 (2, 'Tanisha', 102, 'Finance', 'Female', 'Mumbai'),  
13 (3, 'Varshitha', 103, 'Marketing', 'Female', 'Delhi'),  
14 (4, 'Neha', 104, 'Sales', 'Female', 'Chennai'),  
15 (5, 'Spoorthi', 101, 'IT', 'Female', 'Bengaluru'),  
16 (6, 'Navya', 103, 'Marketing', 'Female', 'Mumbai'),  
17 (7, 'Tom', 106, 'Finance', 'Male', 'Chennai'),  
18 (8, 'Ram', 109, 'Marketing', 'Male', 'Bengaluru');  
19  
20 SELECT worker_name,  
21        SUBSTRING(worker_name, 1, 3) AS Short_Name  
22 FROM workers;  
23
```

Output:

worker_name	Short_Name
Manya	Man
Tanisha	Tan
Varshitha	Var
Neha	Neh
Spoorthi	Spo
Navya	Nav
Tom	Tom
Ram	Ram

NUMERIC FUNCTIONS:

❖ ROUND():



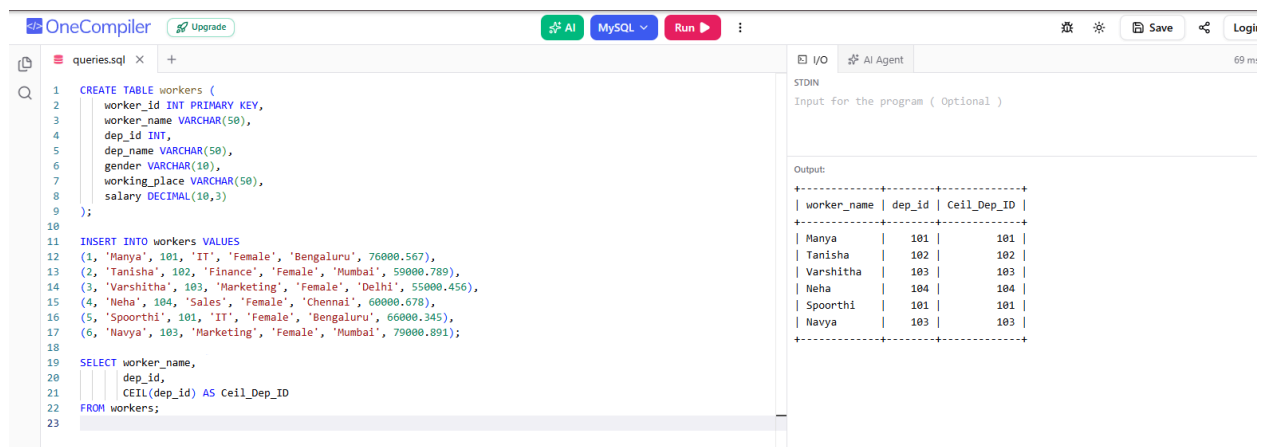
The screenshot shows the OneCompiler MySQL interface. The query editor contains the following SQL code:

```
1 CREATE TABLE workers (  
2     worker_id INT PRIMARY KEY,  
3     worker_name VARCHAR(50),  
4     dep_id INT,  
5     dep_name VARCHAR(50),  
6     gender VARCHAR(10),  
7     working_place VARCHAR(50),  
8     salary DECIMAL(10,3)  
9 );  
10  
11 INSERT INTO workers VALUES  
12 (1, 'Many', 101, 'IT', 'Female', 'Bengaluru', 76000.567),  
13 (2, 'Tanisha', 102, 'Finance', 'Female', 'Mumbai', 59000.789),  
14 (3, 'Varshitha', 103, 'Marketing', 'Female', 'Delhi', 55000.456),  
15 (4, 'Neha', 104, 'Sales', 'Female', 'Chennai', 60000.678),  
16 (5, 'Spoorthi', 101, 'IT', 'Female', 'Bengaluru', 66000.345),  
17 (6, 'Navya', 103, 'Marketing', 'Female', 'Mumbai', 79000.891);  
18  
19 SELECT worker_name,  
20         salary,  
21         ROUND(salary, 2) AS Rounded_Salary  
22 FROM workers;  
23  
24
```

The output window shows the result of the query:

worker_name	salary	Rounded_Salary
Many	76000.567	76000.57
Tanisha	59000.789	59000.79
Varshitha	55000.456	55000.46
Neha	60000.678	60000.68
Spoorthi	66000.345	66000.35
Navya	79000.891	79000.89

❖ CEIL():



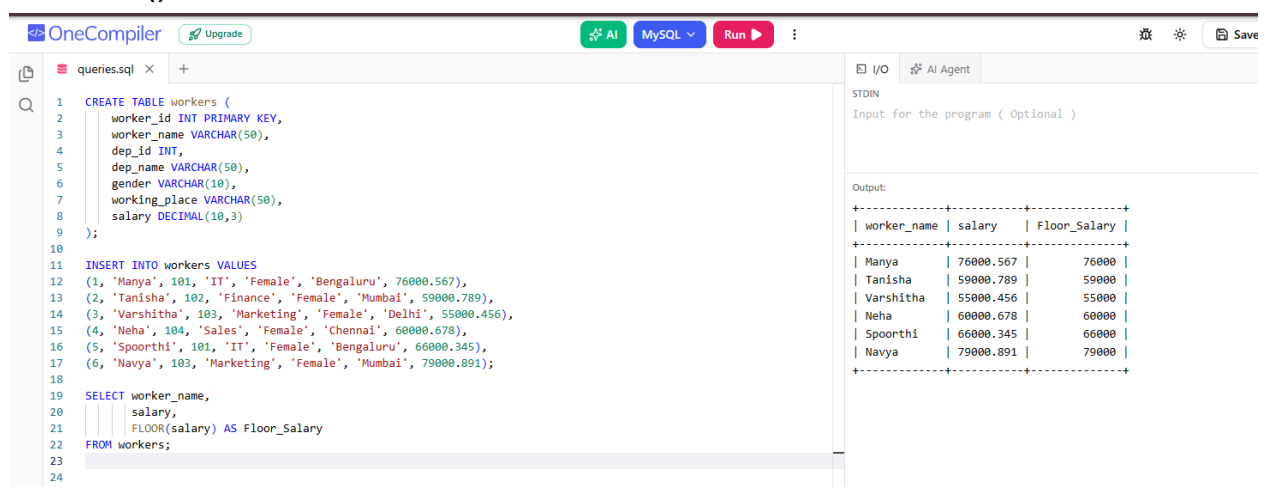
The screenshot shows the OneCompiler MySQL interface. The query editor contains the following SQL code:

```
1 CREATE TABLE workers (  
2     worker_id INT PRIMARY KEY,  
3     worker_name VARCHAR(50),  
4     dep_id INT,  
5     dep_name VARCHAR(50),  
6     gender VARCHAR(10),  
7     working_place VARCHAR(50),  
8     salary DECIMAL(10,3)  
9 );  
10  
11 INSERT INTO workers VALUES  
12 (1, 'Many', 101, 'IT', 'Female', 'Bengaluru', 76000.567),  
13 (2, 'Tanisha', 102, 'Finance', 'Female', 'Mumbai', 59000.789),  
14 (3, 'Varshitha', 103, 'Marketing', 'Female', 'Delhi', 55000.456),  
15 (4, 'Neha', 104, 'Sales', 'Female', 'Chennai', 60000.678),  
16 (5, 'Spoorthi', 101, 'IT', 'Female', 'Bengaluru', 66000.345),  
17 (6, 'Navya', 103, 'Marketing', 'Female', 'Mumbai', 79000.891);  
18  
19 SELECT worker_name,  
20         dep_id,  
21         CEIL(dep_id) AS Ceil_Dep_ID  
22 FROM workers;  
23  
24
```

The output window shows the result of the query:

worker_name	dep_id	Ceil_Dep_ID
Many	101	101
Tanisha	102	102
Varshitha	103	103
Neha	104	104
Spoorthi	101	101
Navya	103	103

❖ FLOOR():



The screenshot shows the OneCompiler MySQL interface. The query editor contains the following SQL code:

```
1 CREATE TABLE workers (  
2     worker_id INT PRIMARY KEY,  
3     worker_name VARCHAR(50),  
4     dep_id INT,  
5     dep_name VARCHAR(50),  
6     gender VARCHAR(10),  
7     working_place VARCHAR(50),  
8     salary DECIMAL(10,3)  
9 );  
10  
11 INSERT INTO workers VALUES  
12 (1, 'Many', 101, 'IT', 'Female', 'Bengaluru', 76000.567),  
13 (2, 'Tanisha', 102, 'Finance', 'Female', 'Mumbai', 59000.789),  
14 (3, 'Varshitha', 103, 'Marketing', 'Female', 'Delhi', 55000.456),  
15 (4, 'Neha', 104, 'Sales', 'Female', 'Chennai', 60000.678),  
16 (5, 'Spoorthi', 101, 'IT', 'Female', 'Bengaluru', 66000.345),  
17 (6, 'Navya', 103, 'Marketing', 'Female', 'Mumbai', 79000.891);  
18  
19 SELECT worker_name,  
20         salary,  
21         FLOOR(salary) AS Floor_Salary  
22 FROM workers;  
23  
24
```

The output window shows the result of the query:

worker_name	salary	Floor_Salary
Many	76000.567	76000
Tanisha	59000.789	59000
Varshitha	55000.456	55000
Neha	60000.678	60000
Spoorthi	66000.345	66000
Navya	79000.891	79000

❖ MOD():

The screenshot shows the OneCompiler MySQL IDE. The query editor contains the following SQL code:

```
1 CREATE TABLE workers (  
2     worker_id INT PRIMARY KEY,  
3     worker_name VARCHAR(50),  
4     dep_id INT,  
5     dep_name VARCHAR(50),  
6     gender VARCHAR(10),  
7     working_place VARCHAR(50),  
8     salary DECIMAL(10,3)  
9 );  
10  
11 INSERT INTO workers VALUES  
12 (1, 'Manya', 101, 'IT', 'Female', 'Bengaluru', 76000.567),  
13 (2, 'Tanisha', 102, 'Finance', 'Female', 'Mumbai', 59000.789),  
14 (3, 'Varshitha', 103, 'Marketing', 'Female', 'Delhi', 55000.456),  
15 (4, 'Neha', 104, 'Sales', 'Female', 'Chennai', 60000.678),  
16 (5, 'Spoorthi', 101, 'IT', 'Female', 'Bengaluru', 66000.345),  
17 (6, 'Navya', 103, 'Marketing', 'Female', 'Mumbai', 79000.891);  
18  
19 SELECT worker_name,  
20        salary,  
21        MOD(salary, 1000) AS Salary_Remainder  
22 FROM workers;  
23  
24
```

The output panel shows the result of the query:

worker_name	salary	Salary_Remainder
Manya	76000.567	0.567
Tanisha	59000.789	0.789
Varshitha	55000.456	0.456
Neha	60000.678	0.678
Spoorthi	66000.345	0.345
Navya	79000.891	0.891

❖ ABS():

The screenshot shows the OneCompiler MySQL IDE. The query editor contains the following SQL code:

```
1 CREATE TABLE workers (  
2     worker_id INT PRIMARY KEY,  
3     worker_name VARCHAR(50),  
4     dep_id INT,  
5     dep_name VARCHAR(50),  
6     gender VARCHAR(10),  
7     working_place VARCHAR(50),  
8     salary DECIMAL(10,3)  
9 );  
10  
11 INSERT INTO workers VALUES  
12 (1, 'Manya', 101, 'IT', 'Female', 'Bengaluru', 76000.567),  
13 (2, 'Tanisha', 102, 'Finance', 'Female', 'Mumbai', 59000.789),  
14 (3, 'Varshitha', 103, 'Marketing', 'Female', 'Delhi', 55000.456),  
15 (4, 'Neha', 104, 'Sales', 'Female', 'Chennai', 60000.678),  
16 (5, 'Spoorthi', 101, 'IT', 'Female', 'Bengaluru', 66000.345),  
17 (6, 'Navya', 103, 'Marketing', 'Female', 'Mumbai', 79000.891);  
18  
19 SELECT worker_name,  
20        salary,  
21        ABS(salary) AS Absolute_Salary  
22 FROM workers;  
23
```

The output panel shows the result of the query:

worker_name	salary	Absolute_Salary
Manya	76000.567	76000.567
Tanisha	59000.789	59000.789
Varshitha	55000.456	55000.456
Neha	60000.678	60000.678
Spoorthi	66000.345	66000.345
Navya	79000.891	79000.891

DATE FUNCTIONS:

- Extracting the year:

queries.sql × +

```
1 CREATE TABLE STUDENT (  
2     Student_ID INT PRIMARY KEY,  
3     Student_Name VARCHAR(50),  
4     Date_of_Birth DATE  
5 );  
6 INSERT INTO STUDENT VALUES  
7 (1, 'Many', '2005-05-15'),  
8 (2, 'Tanish', '2004-08-20'),  
9 (3, 'Varshitha', '2005-01-10'),  
10 (4, 'Neha', '2004-11-25'),  
11 (5, 'Spoorthi', '2005-03-18'),  
12 (6, 'Navya', '2004-07-12');  
13 SELECT * FROM STUDENT;  
14 SELECT Student_Name,  
15         Date_of_Birth,  
16         YEAR(Date_of_Birth) AS Birth_Year  
17 FROM STUDENT;
```

I/O AI Agent

STDIN
Input for the program (Optional)

Output:

Student_ID	Student_Name	Date_of_Birth
1	Many	2005-05-15
2	Tanish	2004-08-20
3	Varshitha	2005-01-10
4	Neha	2004-11-25
5	Spoorthi	2005-03-18
6	Navya	2004-07-12

Student_Name	Date_of_Birth	Birth_Year
Many	2005-05-15	2005
Tanish	2004-08-20	2004
Varshitha	2005-01-10	2005
Neha	2004-11-25	2004
Spoorthi	2005-03-18	2005
Navya	2004-07-12	2004

- Extracting the Month:

queries.sql × +

```
1 CREATE TABLE STUDENT (  
2     Student_ID INT PRIMARY KEY,  
3     Student_Name VARCHAR(50),  
4     Date_of_Birth DATE  
5 );  
6 INSERT INTO STUDENT VALUES  
7 (1, 'Many', '2005-05-15'),  
8 (2, 'Tanish', '2004-08-20'),  
9 (3, 'Varshitha', '2005-01-10'),  
10 (4, 'Neha', '2004-11-25'),  
11 (5, 'Spoorthi', '2005-03-18'),  
12 (6, 'Navya', '2004-07-12');  
13 SELECT * FROM STUDENT;  
14 SELECT Student_Name,  
15         Date_of_Birth,  
16         MONTH(Date_of_Birth) AS Birth_Month  
17 FROM STUDENT;
```

I/O AI Agent

STDIN
Input for the program (Optional)

Output:

Student_ID	Student_Name	Date_of_Birth
1	Many	2005-05-15
2	Tanish	2004-08-20
3	Varshitha	2005-01-10
4	Neha	2004-11-25
5	Spoorthi	2005-03-18
6	Navya	2004-07-12

Student_Name	Date_of_Birth	Birth_Month
Many	2005-05-15	5
Tanish	2004-08-20	8
Varshitha	2005-01-10	1
Neha	2004-11-25	11
Spoorthi	2005-03-18	3
Navya	2004-07-12	7

- Extracting the day:

queries.sql

+

I/O

AI Agent

```

1 CREATE TABLE STUDENT (
2     Student_ID INT PRIMARY KEY,
3     Student_Name VARCHAR(50),
4     Date_of_Birth DATE
5 );
6 INSERT INTO STUDENT VALUES
7 (1, 'Many', '2005-05-15'),
8 (2, 'Tanish', '2004-08-20'),
9 (3, 'Varshitha', '2005-01-10'),
10 (4, 'Neha', '2004-11-25'),
11 (5, 'Spoorthi', '2005-03-18'),
12 (6, 'Navya', '2004-07-12');
13 SELECT * FROM STUDENT;
14 SELECT Student_Name,
15         Date_of_Birth,
16         DAY(Date_of_Birth) AS Birth_Day
17 FROM STUDENT;

```

STDIN

Input for the program (Optional)

Output:

Student_ID	Student_Name	Date_of_Birth
1	Many	2005-05-15
2	Tanish	2004-08-20
3	Varshitha	2005-01-10
4	Neha	2004-11-25
5	Spoorthi	2005-03-18
6	Navya	2004-07-12

Student_Name	Date_of_Birth	Birth_Day
Many	2005-05-15	15
Tanish	2004-08-20	20
Varshitha	2005-01-10	10
Neha	2004-11-25	25
Spoorthi	2005-03-18	18
Navya	2004-07-12	12

- Calculating Approximate Age

queries.sql

+

I/O

AI Agent

```

1 CREATE TABLE STUDENT (
2     Student_ID INT PRIMARY KEY,
3     Student_Name VARCHAR(50),
4     Date_of_Birth DATE
5 );
6 INSERT INTO STUDENT VALUES
7 (1, 'Many', '2005-05-15'),
8 (2, 'Tanish', '2004-08-20'),
9 (3, 'Varshitha', '2005-01-10'),
10 (4, 'Neha', '2004-11-25'),
11 (5, 'Spoorthi', '2005-03-18'),
12 (6, 'Navya', '2004-07-12');
13 SELECT * FROM STUDENT;
14 SELECT Student_Name,
15         Date_of_Birth,
16         TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE()) AS Age
17 FROM STUDENT;

```

STDIN

Input for the program (Optional)

Output:

Student_ID	Student_Name	Date_of_Birth
1	Many	2005-05-15
2	Tanish	2004-08-20
3	Varshitha	2005-01-10
4	Neha	2004-11-25
5	Spoorthi	2005-03-18
6	Navya	2004-07-12

Student_Name	Date_of_Birth	Age
Many	2005-05-15	21
Tanish	2004-08-20	22
Varshitha	2005-01-10	21
Neha	2004-11-25	21
Spoorthi	2005-03-18	21
Navya	2004-07-12	22

- CURDATE() :

queries.sql

+

```
1 CREATE TABLE STUDENT (
2     Student_ID INT PRIMARY KEY,
3     Student_Name VARCHAR(50),
4     Date_of_Birth DATE
5 );
6 INSERT INTO STUDENT VALUES
7 (1, 'Many', '2005-05-15'),
8 (2, 'Tanish', '2004-08-20'),
9 (3, 'Varshitha', '2005-01-10'),
10 (4, 'Neha', '2004-11-25'),
11 (5, 'Spoorthi', '2005-03-18'),
12 (6, 'Navya', '2004-07-12');
13 SELECT * FROM STUDENT;
14 SELECT CURDATE() AS Today;
```

I/O

AI Agent

STDIN

Input for the program (Optional)

Output:

Student_ID	Student_Name	Date_of_Birth
1	Many	2005-05-15
2	Tanish	2004-08-20
3	Varshitha	2005-01-10
4	Neha	2004-11-25
5	Spoorthi	2005-03-18
6	Navya	2004-07-12

Today
2026-09-01

AGGREGATE FUNCTIONS:

- COUNT() – Count Students

queries.sql

+

```
1 CREATE TABLE STUDENT (
2     Student_ID INT PRIMARY KEY,
3     Student_Name VARCHAR(50),
4     Date_of_Birth DATE
5 );
6 INSERT INTO STUDENT VALUES
7 (1, 'Many', '2005-05-15'),
8 (2, 'Tanish', '2004-08-20'),
9 (3, 'Varshitha', '2005-01-10'),
10 (4, 'Neha', '2004-11-25'),
11 (5, 'Spoorthi', '2005-03-18'),
12 (6, 'Navya', '2004-07-12');
13 SELECT * FROM STUDENT;
14 SELECT COUNT(*) AS Total_Students
15 FROM STUDENT;
```

I/O

AI Agent

STDIN

Input for the program (Optional)

Output:

Student_ID	Student_Name	Date_of_Birth
1	Many	2005-05-15
2	Tanish	2004-08-20
3	Varshitha	2005-01-10
4	Neha	2004-11-25
5	Spoorthi	2005-03-18
6	Navya	2004-07-12

Total_Students
6

- SUM() – Calculate Total Marks

queries.sql

+

```
1 CREATE TABLE STUDENT (
2     Student_ID INT PRIMARY KEY,
3     Student_Name VARCHAR(50),
4     Date_of_Birth DATE,
5     Marks INT
6 );
7 INSERT INTO STUDENT VALUES
8 (1, 'Many', '2005-05-15', 85),
9 (2, 'Tanish', '2004-08-20', 78),
10 (3, 'Varshitha', '2005-01-10', 92),
11 (4, 'Neha', '2004-11-25', 74),
12 (5, 'Spoorthi', '2005-03-18', 88),
13 (6, 'Navya', '2004-07-12', 81);
14 SELECT * FROM STUDENT;
15 SELECT SUM(Marks) AS Total_Marks
16 FROM STUDENT;
```

I/O

AI Agent

STDIN

Input for the program (Optional)

Output

Student_ID	Student_Name	Date_of_Birth	Marks
1	Many	2005-05-15	85
2	Tanish	2004-08-20	78
3	Varshitha	2005-01-10	92
4	Neha	2004-11-25	74
5	Spoorthi	2005-03-18	88
6	Navya	2004-07-12	81

Total_Marks
498

- AVG()– Calculate Average Marks

queries.sql X +

```
1 CREATE TABLE STUDENT (  
2     Student_ID INT PRIMARY KEY,  
3     Student_Name VARCHAR(50),  
4     Date_of_Birth DATE,  
5     Marks INT  
6 );  
7 INSERT INTO STUDENT VALUES  
8 (1, 'Many', '2005-05-15', 85),  
9 (2, 'Tanish', '2004-08-20', 78),  
10 (3, 'Varshitha', '2005-01-10', 92),  
11 (4, 'Neha', '2004-11-25', 74),  
12 (5, 'Spoorthi', '2005-03-18', 88),  
13 (6, 'Navya', '2004-07-12', 81);  
14 SELECT * FROM STUDENT;  
15 SELECT AVG(Marks) AS Average_Marks  
16 FROM STUDENT;
```

I/O AI Agent

STDIN
Input for the program (Optional)

Output:

Student_ID	Student_Name	Date_of_Birth	Marks
1	Many	2005-05-15	85
2	Tanish	2004-08-20	78
3	Varshitha	2005-01-10	92
4	Neha	2004-11-25	74
5	Spoorthi	2005-03-18	88
6	Navya	2004-07-12	81

Average_Marks
83.0000

- MAX() – Find Highest Marks

queries.sql

+

```
1 CREATE TABLE STUDENT (
2     Student_ID INT PRIMARY KEY,
3     Student_Name VARCHAR(50),
4     Date_of_Birth DATE,
5     Marks INT
6 );
7 INSERT INTO STUDENT VALUES
8 (1, 'Many', '2005-05-15', 85),
9 (2, 'Tanish', '2004-08-20', 78),
10 (3, 'Varshitha', '2005-01-10', 92),
11 (4, 'Neha', '2004-11-25', 74),
12 (5, 'Spoorthi', '2005-03-18', 88),
13 (6, 'Navya', '2004-07-12', 81);
14 SELECT * FROM STUDENT;
15 SELECT MAX(Marks) AS Highest_Marks
16 FROM STUDENT;
```

I/O

AI Agent

STDIN

Input for the program (Optional)

Output:

Student_ID	Student_Name	Date_of_Birth	Marks
1	Many	2005-05-15	85
2	Tanish	2004-08-20	78
3	Varshitha	2005-01-10	92
4	Neha	2004-11-25	74
5	Spoorthi	2005-03-18	88
6	Navya	2004-07-12	81

Highest_Marks
92

- **MIN() - Find Lowest Marks**

queries.sql

+

I/O

AI Agent

STDIN

Input for the program (Optional)

```

1 CREATE TABLE STUDENT (
2     Student_ID INT PRIMARY KEY,
3     Student_Name VARCHAR(50),
4     Date_of_Birth DATE,
5     Marks INT
6 );
7 INSERT INTO STUDENT VALUES
8 (1, 'Manya', '2005-05-15', 85),
9 (2, 'Tanish', '2004-08-20', 78),
10 (3, 'Varshitha', '2005-01-10', 92),
11 (4, 'Neha', '2004-11-25', 74),
12 (5, 'Spoorthi', '2005-03-18', 88),
13 (6, 'Navya', '2004-07-12', 81);
14 SELECT * FROM STUDENT;
15 SELECT MIN(Marks) AS Lowest_Marks
16 FROM STUDENT;

```

Output:

Student_ID	Student_Name	Date_of_Birth	Marks
1	Manya	2005-05-15	85
2	Tanish	2004-08-20	78
3	Varshitha	2005-01-10	92
4	Neha	2004-11-25	74
5	Spoorthi	2005-03-18	88
6	Navya	2004-07-12	81

Lowest_Marks
74

Aggregate Functions with:

- **GROUP BY :**

queries.sql

+

I/O

AI Agent

STDIN

Input for the program (Optional)

```

1 CREATE TABLE STUDENTS (
2     student_usn VARCHAR(20) PRIMARY KEY,
3     student_name VARCHAR(60),
4     student_age INT,
5     student_email VARCHAR(60),
6     student_marks INT,
7     student_class VARCHAR(20)
8 );
9 INSERT INTO STUDENTS VALUES
10 ('USN001', 'Manya', 21, 'many@gmail.com', 85, '10A'),
11 ('USN002', 'Tanisha', 22, 'tanish@gmail.com', 78, '10A'),
12 ('USN003', 'Varshitha', 21, 'varshitha@gmail.com', 92, '10B'),
13 ('USN004', 'Neha', 22, 'neha@gmail.com', 74, '10B'),
14 ('USN005', 'Spoorthi', 21, 'spoorthi@gmail.com', 88, '10A'),
15 ('USN006', 'Navya', 22, 'navya@gmail.com', 81, '10B');
16 SELECT * FROM STUDENTS
17 ORDER BY student_marks DESC;

```

Output:

student_usn	student_name	student_age	student_email	student_marks	student_class
USN003	Varshitha	21	varshitha@gmail.com	92	10B
USN005	Spoorthi	21	spoorthi@gmail.com	88	10A
USN001	Manya	21	many@gmail.com	85	10A
USN006	Navya	22	navya@gmail.com	81	10B
USN002	Tanisha	22	tanish@gmail.com	78	10A
USN004	Neha	22	neha@gmail.com	74	10B

- ORDER BY:

```
queries.sql × +
1 CREATE TABLE STUDENTS (
2     student_usn VARCHAR(20) PRIMARY KEY,
3     student_name VARCHAR(60),
4     student_age INT,
5     student_email VARCHAR(60),
6     student_marks INT,
7     student_class VARCHAR(20)
8 );
9 INSERT INTO STUDENTS VALUES
10 ('USN001', 'Manu', 21, 'many@gmail.com', 85, '10A'),
11 ('USN002', 'Tanisha', 22, 'tanish@gmail.com', 78, '10A'),
12 ('USN003', 'Varshitha', 21, 'varshitha@gmail.com', 92, '10B'),
13 ('USN004', 'Neha', 22, 'neha@gmail.com', 74, '10B'),
14 ('USN005', 'Spoorthi', 21, 'spoorthi@gmail.com', 88, '10A'),
15 ('USN006', 'Navya', 22, 'navya@gmail.com', 81, '10B');
16 SELECT * FROM STUDENTS;
17 SELECT student_class AS Class,
18        COUNT(*) AS Total_Students,
19        AVG(student_marks) AS Average_Marks
20 FROM STUDENTS
21 GROUP BY student_class;
22
```

OUTPUT:

Output:

student_usn	student_name	student_age	student_email	student_marks	student_class
USN001	Manu	21	many@gmail.com	85	10A
USN002	Tanisha	22	tanish@gmail.com	78	10A
USN003	Varshitha	21	varshitha@gmail.com	92	10B
USN004	Neha	22	neha@gmail.com	74	10B
USN005	Spoorthi	21	spoorthi@gmail.com	88	10A
USN006	Navya	22	navya@gmail.com	81	10B

Class	Total_Students	Average_Marks
10A	3	83.6667
10B	3	82.3333

GIHUB LINK:

